



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

INTRODUCTION TO COMPUTER ENGINEERING: PROGRAMMING AND APPLICATIONS

TUTORIAL 2

DATA REPRESENTATION AND CONVERSION

Outline

- 1. Data representation using positional notation.
 - Conversion from binary(2), octal(8) and hexadecimal(16) system to decimal(10) system.
- 2. Conversion from decimal system(10) to hexadecimal(16), octal(8) and binary(2) system.
 - i. Integer part.
 - ii. Fractional part.
- 3. One-to-one relationship between i) octal and binary system; ii) hexadecimal and binary system.

Number System

- The basic idea of **positional notation**: Each number system contains two elements, a **base** and a set of **symbols**. Symbols put at different positions have different value(weight). For example, $(1901.56)_{10}$, $(1001.1001)_2$, $(123.45)_8$, $(4ED.C1)_{16}$ represent numbers in different number system, and they can be decomposed into a **series** as shown in the next page. For hexadecimal system: **$A := 10$, $B := 11$, $C := 12$, $D := 13$, $E := 14$, $F := 15$.**

Number system	Base	Set of symbols
Decimal system	10	0,1,2,3,4,5,6,7,8,9
Binary system	2	0,1
Octal system	8	0,1,2,3,4,5,6,7
Hexadecimal system	16	0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

Data Representation

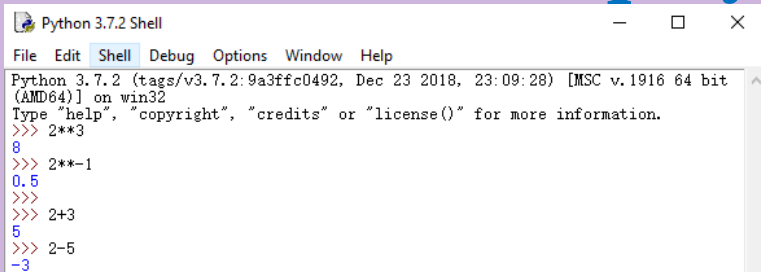
- **Data Decomposition**(The first symbol at the left of decimal point “.” has position $i = 0$, and increase($i += 1$) towards left while decrease($i -= 1$) toward right.):

$$N = a_n \times b^n + a_{n-1} \times b^{n-1} + a_{n-2} \times b^{n-2} \dots + a_0 \times b^0 \text{ (Integer part)} \\ + a_{-1} \times b^{-1} + a_{-2} \times b^{-2} \dots \text{ (Fractional part)}$$

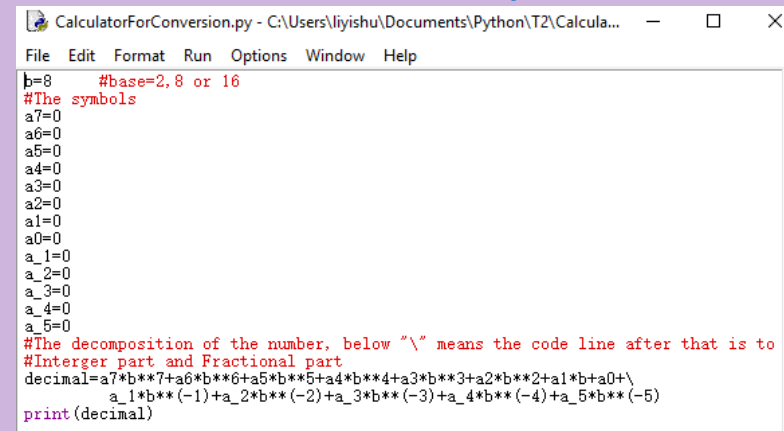
- Every number can be decomposed into the sum of a series of numbers, each is represented by a set of positional value($\{a_i | i \in \mathbb{Z}\}$) times the corresponding weight($\{b^i | i \in \mathbb{Z}\}$), where
 - For decimal system: $b=10$, $a_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.
 - For binary system: $b=2$, $a_i \in \{0, 1\}$.
 - For octal system: $b=8$, $a_i \in \{0, 1, 2, 3, 4, 5, 6, 7\}$.
 - For hexadecimal system: $b=16$, $a_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F\}$, where $A := 10$, $B := 11$, $C := 12$, $D := 13$, $E := 14$, $F := 15$.

Use Python as calculator

- In Python Shell, you could just type the computational formulas and press Enter, the result will be auto calculated and displayed.



```
Python 3.7.2 Shell
File Edit Shell Debug Options Window Help
Python 3.7.2 (tags/v3.7.2:9a3ffc0492, Dec 23 2018, 23:09:28) [MSC v.1916 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> 2**3
8
>>> 2**-1
0.5
>>>
>>> 2+3
5
>>> 2-5
-3
```



```
CalculatorForConversion.py - C:\Users\liyishu\Documents\Python\T2\Calcula...
File Edit Format Run Options Window Help
b=8          #base=2, 8 or 16
#The symbols
a7=0
a6=0
a5=0
a4=0
a3=0
a2=0
a1=0
a0=0
a_1=0
a_2=0
a_3=0
a_4=0
a_5=0
#The decomposition of the number, below "\n" means the code line after that is to
#Integer part and Fractional part
decimal=a7*b**7+a6*b**6+a5*b**5+a4*b**4+a3*b**3+a2*b**2+a1*b+a0+\
        a_1*b**(-1)+a_2*b**(-2)+a_3*b**(-3)+a_4*b**(-4)+a_5*b**(-5)
print(decimal)
```

- Create a new py file, define different variables, do calculation for these variables, and print the result.

Practice I

- i) Binary number 100.1 equals to decimal number 4.5
- ii) Octal number 100.1 equals to decimal number 64.125
- iii) Hexadecimal number 100.1 equals to decimal number 256.0625

Solution

- $(100.1)_2 = 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} = (4.5)_{10}$
- $(100.1)_8 = 1 \times 8^2 + 0 \times 8^1 + 0 \times 8^0 + 1 \times 8^{-1} = (64.125)_{10}$
- $(100.1)_{16} = 1 \times 16^2 + 0 \times 16^1 + 0 \times 16^0 + 1 \times 16^{-1} = (256.0625)_{10}$

```
>>> 1*2**2+1*2**-1
4.5
>>> 1*8**2+1*8**-1
64.125
>>> 1*16**2+1*16**-1
256.0625
```

```
CalculatorForConversion.py - C:\Users\liyishu\Documents\Python\T2\Calcula...
File Edit Format Run Options Window Help
b=2      #base=2, 8 or 16
#The symbols
a7=0
a6=0
a5=0
a4=0
a3=0
a2=1
a1=0
a0=0
a_1=1
a_2=0
a_3=0
a_4=0
a_5=0
#The decomposition of the number, below "\" means the code line after that is to
#Integer part and Fractional part
decimal=a7*b**7+a6*b**6+a5*b**5+a4*b**4+a3*b**3+a2*b**2+a1*b+a0+\
a_1*b**(-1)+a_2*b**(-2)+a_3*b**(-3)+a_4*b**(-4)+a_5*b**(-5)
print(decimal)

Python 3.7.2 Shell
File Edit Shell Debug Options Window Help
Python 3.7.2 (tags/v3.7.2:9a3ffc0492, Dec 23 2018, 23:09:28) [MSC v.1916 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:\Users\liyishu\Documents\Python\T2\CalculatorForConversion.py ==
4.5
```

```
CalculatorForConversion.py - C:\Users\liyishu\Documents\Python\T2\Calcula...
File Edit Format Run Options Window Help
b=8      #base=2, 8 or 16
#The symbols
a7=0
a6=0
a5=0
a4=0
a3=0
a2=1
a1=0
a0=0
a_1=1
a_2=0
a_3=0
a_4=0
a_5=0
#The decomposition of the number, below "\" means the code line after that is to
#Integer part and Fractional part
decimal=a7*b**7+a6*b**6+a5*b**5+a4*b**4+a3*b**3+a2*b**2+a1*b+a0+\
a_1*b**(-1)+a_2*b**(-2)+a_3*b**(-3)+a_4*b**(-4)+a_5*b**(-5)
print(decimal)

Python 3.7.2 Shell
File Edit Shell Debug Options Window Help
Python 3.7.2 (tags/v3.7.2:9a3ffc0492, Dec 23 2018, 23:09:28) [MSC v.1916 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:\Users\liyishu\Documents\Python\T2\CalculatorForConversion.py ==
64.125
```

```
CalculatorForConversion.py - C:\Users\liyishu\Documents\Python\T2\Calcula...
File Edit Format Run Options Window Help
b=16     #base=2, 8 or 16
#The symbols
a7=0
a6=0
a5=0
a4=0
a3=0
a2=1
a1=0
a0=0
a_1=1
a_2=0
a_3=0
a_4=0
a_5=0
#The decomposition of the number, below "\" means the code line after that is to
#Integer part and Fractional part
decimal=a7*b**7+a6*b**6+a5*b**5+a4*b**4+a3*b**3+a2*b**2+a1*b+a0+\
a_1*b**(-1)+a_2*b**(-2)+a_3*b**(-3)+a_4*b**(-4)+a_5*b**(-5)
print(decimal)

Python 3.7.2 Shell
File Edit Shell Debug Options Window Help
Python 3.7.2 (tags/v3.7.2:9a3ffc0492, Dec 23 2018, 23:09:28) [MSC v.1916 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:\Users\liyishu\Documents\Python\T2\CalculatorForConversion.py ==
256.0625
```

Practice 2

- Convert binary number 10101010.0101 into decimal number.
- Convert octal number 6.6, 66.66, 666.666, and 6666.6666 into decimal number.
- Convert hexadecimal number 3D5.F9 into decimal number.

Solution

- $(10101010.0101)_2 = 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4} =$
 $(170.3125)_{10}$
- $(6.6)_8 = 6 \times 8^0 + 6 \times 8^{-1} = (6.75)_{10}$
 $(66.66)_8 = 6 \times 8^1 + 6 \times 8^0 + 6 \times 8^{-1} + 6 \times 8^{-2} = (54.84375)_{10}$
 $(666.666)_8 = 6 \times 8^2 + 6 \times 8^1 + 6 \times 8^0 + 6 \times 8^{-1} + 6 \times 8^{-2} + 6 \times 8^{-3} =$
 $(438.85546875)_{10}$
 $(6666.6666)_8 = 6 \times 8^3 + 6 \times 8^2 + 6 \times 8^1 + 6 \times 8^0 + 6 \times 8^{-1} + 6 \times 8^{-2} + 6 \times 8^{-3} + 6 \times 8^{-4} = (3510.85693359375)_{10}$
- $(3D5.F9)_{16} = 3 \times 16^2 + 13 \times 16^1 + 5 \times 16^0 + 15 \times 16^{-1} + 9 \times 16^{-2} =$
 $(981.97265625)_{10}$

From Decimal to Others

- For integer part, use **division** by the bases. Because $N = a_n \times b^n + a_{n-1} \times b^{n-1} + a_{n-2} \times b^{n-2} \dots + a_0 \times b^0$ (**Integer part**), so
- i) $\text{mod}(N, b) = a_0$
 - ii) $\text{mod}(N/b, b) = a_1$
 - iii) $\text{mod}(N/b/b, b) = a_2 \dots$ until finally $\text{mod}(a_n \times b, b) = a_n$.
- So the representation for $N = (a_n a_{n-1} \dots a_0)_b$.
- For fractional part, use **multiplication** by the bases. Because $N = 0 + a_{-1} \times b^{-1} + a_{-2} \times b^{-2} + \dots$ (**Fractional part**), so
- i) $N \times b = a_{-1} + a_{-2} \times b^{-1} + a_{-3} \times b^{-2} + \dots$
 - ii) $(N \times b - a_{-1}) \times b = a_{-2} + a_{-3} \times b^{-1} + a_{-4} \times b^{-2} + \dots$
 - iii) ... until finally a_{-n} left, no negative power terms of b .
- Then the representation for $N = (0. a_{-1} a_{-2} \dots a_{-n})_b$.

Practice 3

➤ Year 2019 can also be represented as

Year ($7E3$)₁₆

Year (3743)₈

Year (1111100011)₂

Solution

16	2019		3	Lower position
16	126		14 (E)	↑ (2019) ₁₀ = (7E3) ₁₆
	7		7	
Quotient			Remainder	Higher position

8	2019		3	Lower position
8	252		4	↑ (2019) ₁₀ = (3743) ₈
8	31		7	
	3		3	
Quotient			Remainder	Higher position

2	2019		1	Lower position
2	1009		1	↑ (2019) ₁₀ = (11111100011) ₂
2	504		0	
2	252		0	
2	126		0	
2	63		1	
2	31		1	
2	15		1	
2	7		1	
2	3		1	
	1		1	
Quotient			Remainder	Higher position

Practice 4

- Convert decimal number 100.96875 into binary, octal and hexadecimal number.

Binary 1100100.11111

Octal 144.76

Hexadecimal 64.F8

Solution

Integer part:	Fractional part:
$\begin{array}{r} 16 \overline{) 100} \dots\dots\dots 4 \uparrow \\ \underline{6} \dots\dots\dots 6 \end{array}$	$\begin{array}{r} 0.96875 \times 16 = 15.5 \dots\dots 15 \text{ Higher position} \\ 0.5 \times 16 = 8 \dots\dots\dots 8 \text{ Lower position} \end{array}$
$\begin{array}{r} 8 \overline{) 100} \dots\dots\dots 4 \uparrow \\ 8 \overline{) 12} \dots\dots\dots 4 \\ \underline{1} \dots\dots\dots 1 \end{array}$	$\begin{array}{r} 0.96875 \times 8 = 7.75 \dots\dots\dots 7 \text{ Higher position} \\ 0.75 \times 8 = 6 \dots\dots\dots 6 \text{ Lower position} \end{array}$
$\begin{array}{r} 2 \overline{) 100} \dots\dots\dots 0 \uparrow \\ 2 \overline{) 50} \dots\dots\dots 0 \\ 2 \overline{) 25} \dots\dots\dots 1 \\ 2 \overline{) 12} \dots\dots\dots 0 \\ 2 \overline{) 6} \dots\dots\dots 0 \\ 2 \overline{) 3} \dots\dots\dots 1 \\ \underline{1} \dots\dots\dots 1 \end{array}$	$\begin{array}{r} 0.96875 \times 2 = 1.9375 \dots\dots 1 \text{ Higher position} \\ 0.9375 \times 2 = 1.875 \dots\dots\dots 1 \\ 0.875 \times 2 = 1.75 \dots\dots\dots 1 \\ 0.75 \times 2 = 1.5 \dots\dots\dots 1 \\ 0.5 \times 2 = 1.0 \dots\dots\dots 1 \text{ Lower position} \end{array}$

$$(100.96875)_{10} = (64.F8)_{16}$$

$$(100.96875)_{10} = (144.76)_8$$

$$(100.96875)_{10} = (1100100.11111)_2$$

Think!

- 1. Convert decimal number 0.3 into octal number.
- 2. Convert decimal number $\frac{1}{3} = 0.\dot{3}$ into octal number.
- 3. Convert decimal number π into octal number.

Solution?

- 1. $(0.3)_{10} == (0.2\dot{3}1\dot{4}\dot{6})_8$ (Check by calculating the limit of the corresponding infinite series.)
- 2. $(0.\dot{3})_{10} = (0.2\dot{5})_8$
- 3. $(\pi)_{10} = (3.1103755242102643 \dots)_8$

```
Python 3.7.2 Shell
File Edit Shell Debug Options Window Help
Python 3.7.2 (tags/v3.7.2:9a3ffc0492, Dec 23 2018
(AMD64)) on win32
Type "help", "copyright", "credits" or "license()
>>>
===== RESTART: C:\Users\liyishu\Documents\Pyt
3 .
1
1
0
3
7
5
5
2
4
2
1
0
2
6
4
3
```

```
pi_octal.py - C:\Users\liyishu\Documents\Python\T2\pi_octal.py (3.7.2)
File Edit Format Run Options Window Help
from math import pi

integerpart=3
fractionalpart=pi-3
print(integerpart, ".")
#1
integerpart=int(fractionalpart*8)
fractionalpart=fractionalpart*8-integerpart
print(integerpart)
#2
integerpart=int(fractionalpart*8)
fractionalpart=fractionalpart*8-integerpart
print(integerpart)
#3
integerpart=int(fractionalpart*8)
fractionalpart=fractionalpart*8-integerpart
print(integerpart)
#4
integerpart=int(fractionalpart*8)
fractionalpart=fractionalpart*8-integerpart
print(integerpart)
#5
integerpart=int(fractionalpart*8)
fractionalpart=fractionalpart*8-integerpart
print(integerpart)
#6
integerpart=int(fractionalpart*8)
fractionalpart=fractionalpart*8-integerpart
print(integerpart)
#7
integerpart=int(fractionalpart*8)
fractionalpart=fractionalpart*8-integerpart
print(integerpart)
#8
integerpart=int(fractionalpart*8)
fractionalpart=fractionalpart*8-integerpart
print(integerpart)
#9
```


Conversion Between Binary, Octal and Hexadecimal

- Use one-to-one relationship between i) 3-digit binary number and the symbols of octal number, ii) 4-digit binary number and the symbols of hexadecimal number.
- To do conversion between octal and hexadecimal, use binary as a bridge.

Binary and Octal

- The one-to-one relationship between 3-digit binary number and the symbols of octal number?

$$(0)_8 = (000)_2$$

$$(1)_8 = (001)_2$$

$$(2)_8 = (010)_2$$

$$(3)_8 = (011)_2$$

$$(4)_8 = (100)_2$$

$$(5)_8 = (101)_2$$

$$(6)_8 = (110)_2$$

$$(7)_8 = (111)_2$$

Binary and Hexadecimal

- The one-to-one relationship between 4-digit binary number and the symbols of hexadecimal number?

$$(0)_{16} = (0000)_2$$

$$(1)_{16} = (0001)_2$$

$$(2)_{16} = (0010)_2$$

$$(3)_{16} = (0011)_2$$

$$(4)_{16} = (0100)_2$$

$$(5)_{16} = (0101)_2$$

$$(6)_{16} = (0110)_2$$

$$(7)_{16} = (0111)_2$$

$$(8)_{16} = (1000)_2$$

$$(9)_{16} = (1001)_2$$

$$(A)_{16} = (1010)_2$$

$$(B)_{16} = (1011)_2$$

$$(C)_{16} = (1100)_2$$

$$(D)_{16} = (1101)_2$$

$$(E)_{16} = (1110)_2$$

$$(F)_{16} = (1111)_2$$

Practice 5

A 5 . 5 8

2 4 5 . 2 6

- 1. Convert binary number 10100101.01011 to octal and hexadecimal number respectively.
- 2. Convert octal number 7654.123 to hexadecimal number.

Binary 11110101100.001010011

→ F A C . 2 9 8

Solution

- 1.i) $(010100101.010110)_2 = (010100101.010110)_2 = (245.26)_8$
ii) $(10100101.01011)_2 = (10100101.01011000)_2 = (A5.58)_{16}$
- 2. $(7654.123)_8 = (111110101100.001010011)_2$
 $= (111110101100.001010011000)_2$
 $= (FAC.298)_{16}$

Practice 6

- 1. Binary number 100.001 equals to decimal number ____
A. $2^3 + 2^{-3}$ B. $2^2 + 2^{-2}$ C. $2^3 + 2^{-2}$ **D. $2^2 + 2^{-3}$**
- 2. Binary number 0.110011 equals to decimal number ____
A. $2^{-1} + 2^{-2} + 2^{-4} + 2^{-5}$ ✗
B. 0.63
C. $1 - 2^{-3} - 2^{-5} - 2^{-6}$
D. $1 - 2^{-3} - 2^{-4} - 2^{-6}$
- $2^{-1} \ 2^{-2} \ 2^{-5} \ 2^{-6}$

Solution

- 1. Binary number 100.001 equals to decimal number ____
A. $2^3 + 2^{-3}$ B. $2^2 + 2^{-2}$ C. $2^3 + 2^{-2}$ **D. $2^2 + 2^{-3}$**
- 2. Binary number 0.110011 equals to decimal number ____
A. $2^{-1} + 2^{-2} + 2^{-4} + 2^{-5}$
B. 0.63
C. $1 - 2^{-3} - 2^{-5} - 2^{-6}$
D. $1 - 2^{-3} - 2^{-4} - 2^{-6}$

$$\begin{array}{r} 0.110011 \\ + 0.001101 \\ \hline 1.000000 \end{array}$$

Practice 7

➤ 1. Which of the following inequalities is correct?

A. $(0.1111)_2 < (0.777)_{10}$ $\times$ $\text{B. } (0.78)_{10} > (0.C)_{16}$ $\checkmark$

C. $(0.6)_{10} \frac{153.6}{256} \geq (0.AB)_{16} \frac{171}{256}$ $\times$ $\text{D. } (0.101)_2 \leq (0.A)_{16}$ $\times$

➤ 2. Which of the following equalities is incorrect?

A. $(0.875)_{10} = (0.E)_{16}$ $\checkmark$ $\text{B. } (11.4)_8 = (9.8)_{16}$ $\checkmark$

C. $(10)_{10} = (1010)_2$ $\checkmark$ $\text{D. } (101111)_2 = (AF)_{16}$ $\times$

$2F$

Solution

➤ 1. Which of the following inequalities is correct?

A. $(0.111)_2 < (0.777)_{10}$ B. $(0.78)_{10} > (0.C)_{16}$

C. $(0.6)_{10} > (0.AB)_{16}$ D. $(0.101)_2 < (0.A)_{16}$

➤ 2. Which of the following equalities is incorrect?

A. $(0.875)_{10} = (0.E)_{16}$ B. $(11.4)_8 = (9.8)_{16}$

C. $(10)_{10} = (1010)_2$ D. $(101111)_2 = (AF)_{16}$